Three Pillars of 3D Incompressible Navier–Stokes Theory: Kato Local Existence, Fujita–Kato Small-Data Globality, and the Beale–Kato–Majda Continuation Criterion (While the Roof Remains Open)

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Abstract

We collect, in a self-contained form, the three classical pillars of the local and conditional theory of the three-dimensional incompressible Navier–Stokes equations on the torus $\mathbb{T}^3$. Pillar I is Kato's local existence theorem together with an explicit lifespan bound for arbitrary divergence-free data in H^s ($s > 5/2$). Pillar II is the Fujita–Kato theorem of global regularity for sufficiently small data in the critical space $\dot{H}^{1/2}$. Pillar III is the Beale–Kato–Majda continuation criterion in its one-directional form: if the vorticity (or strain) supremum is integrable on $[0, T]$, then a maximal Kato solution remains smooth up to time T. These results form the standard rigorous baseline; the question of unconditional global regularity for arbitrary large smooth data (Clay Millennium Problem Statement B) remains open.

1 Introduction

The three-dimensional incompressible Navier–Stokes equations on the torus $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ read

$$\partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} + \nabla p = \nu\Delta\mathbf{u}, \qquad \nabla\cdot\mathbf{u} = 0, \qquad \mathbf{u}(\cdot,0) = \mathbf{u}_0, \tag{1}$$

with $\nu > 0$. Taking the curl yields the vorticity formulation

$$\partial_t \boldsymbol{\omega} + (\mathbf{u}\cdot\nabla)\boldsymbol{\omega} = (\boldsymbol{\omega}\cdot\nabla)\mathbf{u} + \nu\Delta\boldsymbol{\omega}, \qquad \nabla\cdot\boldsymbol{\omega} = 0. \tag{2}$$

The classical theory of these equations rests on three fundamental results, established between 1964 and 1984. We record them here as a compact reference, with proofs expanded to the level of standard energy and commutator estimates. No claim is made that the large-data global-regularity problem is solved.

2 Pillar I: Kato Local Existence and Lifespan Bound

Theorem 2.1 (Kato local existence)**.** *Let $s > 5/2$ and let $\mathbf{u}_0 \in H^s(\mathbb{T}^3)$ be divergence-free. There exist $T_{\mathrm{loc}} > 0$ and a unique solution*

$$\mathbf{u} \in C([0, T_{\mathrm{loc}}); H^s(\mathbb{T}^3)) \cap C^\infty(\mathbb{T}^3 \times (0, T_{\mathrm{loc}}))$$

of (1)*. Moreover the lifespan satisfies*

$$T_{\mathrm{loc}} \geq \frac{c_s}{\|\mathbf{u}_0\|_{H^s}}$$

for a constant $c_s > 0$ depending only on s and dimension.

Proof. We record the three standard pillars of the Kato theory [1].

(a) Full Kato–Ponce commutator bound. Apply $\Lambda^s = (I-\Delta)^{s/2}$ to (1) and take the L^2 pairing with $\Lambda^s \mathbf{u}$. After cancellation of the pressure and the divergence-free constraint one obtains

$$\frac{1}{2}\frac{d}{dt}\|\mathbf{u}\|_{H^s}^2 + \nu\|\nabla\mathbf{u}\|_{H^s}^2 \le C_s\Big(\|\nabla\mathbf{u}\|_{L^\infty}\|\mathbf{u}\|_{H^s} + \|\nabla\mathbf{u}\|_{L^\infty}\|\nabla\mathbf{u}\|_{H^{s-1}}\Big)\|\mathbf{u}\|_{H^s}.$$

For $s > 5/2$ the second summand is absorbed into the first by the Sobolev embedding $H^{s-1} \hookrightarrow L^\infty$ on the gradient factor (or by interpolating $\|\nabla\mathbf{u}\|_{H^{s-1}} \le C\|\mathbf{u}\|_{H^s}$), yielding the closed Riccati inequality

$$\frac{d}{dt}\|\mathbf{u}\|_{H^s} \le C_s\|\mathbf{u}\|_{H^s}^2. \tag{3}$$

(b) Existence via mollified/Galerkin approximations. Let $\mathbf{u}_0^\varepsilon = J_\varepsilon \mathbf{u}_0$ be a family of smooth mollifications (or finite-mode Galerkin projections) of the initial datum, with $\|\mathbf{u}_0^\varepsilon\|_{H^s} \le \|\mathbf{u}_0\|_{H^s}$. The corresponding approximate solutions $\mathbf{u}^\varepsilon$ satisfy (3) with constants independent of ε. Integrating the Riccati inequality on the common interval

$$\big[0, T_{\mathrm{loc}}\big), \qquad T_{\mathrm{loc}} = \frac{1}{2C_s\|\mathbf{u}_0\|_{H^s}},$$

gives a uniform bound $\sup_{t<T_{\mathrm{loc}}}\|\mathbf{u}^\varepsilon(t)\|_{H^s} \le 2\|\mathbf{u}_0\|_{H^s}$. An Aubin–Lions compactness argument (boundedness in $L^\infty_t H^s$ and of $\partial_t \mathbf{u}^\varepsilon$ in a negative Sobolev space) produces a limit $\mathbf{u} \in L^\infty([0,T_{\mathrm{loc}}); H^s)$ solving (1) distributionally; continuity in time with values in H^s follows by the standard energy-identity argument for strong solutions. Parabolic bootstrap then upgrades to $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times (0, T_{\mathrm{loc}}))$.

(c) L^2 uniqueness. If $\mathbf{u}$ and $\mathbf{v}$ are two strong solutions on $[0,T]$ with the same initial datum, set $\mathbf{w} = \mathbf{u} - \mathbf{v}$. Testing the difference of the momentum equations against $\mathbf{w}$ yields

$$\frac{1}{2}\frac{d}{dt}\|\mathbf{w}\|_{L^2}^2 + \nu\|\nabla\mathbf{w}\|_{L^2}^2 = -\int_{\mathbb{T}^3}\big((\mathbf{w}\cdot\nabla)\mathbf{v}\big)\cdot\mathbf{w}\,dV \le \|\nabla\mathbf{v}\|_{L^\infty}\|\mathbf{w}\|_{L^2}^2.$$

Gronwall's lemma with $\mathbf{w}(0) = 0$ forces $\mathbf{w} \equiv 0$. (The gradient factor is that of the second solution $\mathbf{v}$, consistent with the subtraction $\mathbf{u} - \mathbf{v}$.) □

3 Pillar II: Fujita–Kato Small-Data Global Regularity

Theorem 3.1 (Fujita–Kato)**.** *There exists an absolute dimensionless constant $\varepsilon > 0$ such that if $\mathbf{u}_0 \in \dot H^{1/2}(\mathbb{T}^3)$ satisfies*

$$\|\mathbf{u}_0\|_{\dot H^{1/2}} \le \varepsilon\nu,$$

then the solution of Theorem 2.1 exists globally and satisfies

$$\mathbf{u} \in C([0,\infty); \dot H^{1/2}(\mathbb{T}^3)) \cap L^2([0,\infty); \dot H^{3/2}(\mathbb{T}^3)) \cap C^\infty(\mathbb{T}^3 \times (0,\infty)).$$

Proof. Testing (1) against $(-\Delta)^{1/2}\mathbf{u}$ and applying product estimates in homogeneous Sobolev spaces [2] gives

$$\frac{1}{2}\frac{d}{dt}\|\mathbf{u}(t)\|_{\dot H^{1/2}}^2 + \nu\|\mathbf{u}(t)\|_{\dot H^{3/2}}^2 \le C_{\mathrm{FK}}\|\mathbf{u}(t)\|_{\dot H^{1/2}}\|\mathbf{u}(t)\|_{\dot H^{3/2}}^2.$$

Choosing $\varepsilon < 1/(2C_{\mathrm{FK}})$ ensures that $(\nu - C_{\mathrm{FK}}\|\mathbf{u}_0\|_{\dot H^{1/2}}) > \nu/2 > 0$ at $t = 0$, trapping $\|\mathbf{u}(t)\|_{\dot H^{1/2}} \le \|\mathbf{u}_0\|_{\dot H^{1/2}}$ for all $t \ge 0$ and forcing

$$\int_0^\infty \|\mathbf{u}(t)\|_{\dot H^{3/2}}^2\,dt \le \frac{1}{\nu}\|\mathbf{u}_0\|_{\dot H^{1/2}}^2.$$

Interpolating this global $\dot H^{3/2}$ dissipation into higher-order H^s norms yields global control of $\|\nabla\mathbf{u}(t)\|_{L^\infty}$ in $L^1([0,\infty))$. By the local existence theory of Pillar I and the continuation criterion of Pillar III, one obtains $T^* = +\infty$ and smoothness for $t > 0$. □

4 Pillar III: Beale–Kato–Majda Continuation Criterion

Theorem 4.1 (Beale–Kato–Majda, one-directional)**.** *Let* $\mathbf{u}$ *be the maximal Kato solution of Theorem 2.1 with existence time* T^**. If, for a finite* $T \le T^*$*,*

$$\int_0^T \|\boldsymbol{\omega}(t)\|_{L^\infty(\mathbb{T}^3)}\,dt < \infty$$

(or, equivalently, $\int_0^T \|S(t)\|_{L^\infty}\,dt < \infty$ *with* $S = \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^T)$*), then* $T^* > T$*. In particular, if the integral remains finite on every finite horizon then* $T^* = +\infty$ *and the solution is globally smooth for* $t > 0$*.*

Proof. The logarithmic Sobolev inequality of Beale–Kato–Majda (refined by Kozono–Taniuchi) states

$$\|\nabla\mathbf{u}\|_{L^\infty} \le C\Big(1 + \|\boldsymbol{\omega}\|_{L^\infty}\log\big(e + \|\mathbf{u}\|_{H^s}\big) + \|\boldsymbol{\omega}\|_{L^2}\Big).$$

Inserting this bound into the Kato energy inequality

$$\frac{d}{dt}\|\mathbf{u}\|_{H^s}^2 \le C_s\|\nabla\mathbf{u}\|_{L^\infty}\|\mathbf{u}\|_{H^s}^2$$

and setting $Y(t) := \log(e + \|\mathbf{u}\|_{H^s}^2)$ produces

$$\frac{d}{dt}Y(t) \le C'\big(1 + \|\boldsymbol{\omega}(t)\|_{L^2} + \|\boldsymbol{\omega}(t)\|_{L^\infty}\big)Y(t).$$

Gronwall's lemma, together with the unconditional $L_t^2L_x^2$ enstrophy bound from the energy inequality, shows that finiteness of $\int_0^T \|\boldsymbol{\omega}\|_{L^\infty}\,dt$ keeps $\|\mathbf{u}(T)\|_{H^s}$ finite. By the local existence theory of Pillar I the solution can be continued past T. (We state only this one-directional implication; the converse "blow-up forces the integral to diverge" for smooth solutions is classical but is not needed for the continuation applications recorded here.) □

5 Concluding remarks and status matrix

The three theorems together constitute the classical baseline of the local and conditional theory. Pillar I guarantees a unique smooth solution for a short time for every reasonable initial datum. Pillar II shows that the solution is in fact global when the datum is small in the critical norm. Pillar III gives a precise one-directional obstruction to continuation for large data: the time-integrability of the vorticity (or strain) supremum. Whether this obstruction can ever occur for smooth solutions of the Navier–Stokes equations remains the open Clay Millennium Problem.

Table 1: Status matrix for the three classical pillars.

Item	**Status**	**Content**
Pillar I (Kato)	Unconditional Local	Lifespan $T_{\text{loc}} \ge c_s/\|\mathbf{u}_0\|_{H^s}$; smoothness for $t > 0$
Pillar II (Fujita–Kato)	Unconditional Small-Data Global	$\|\mathbf{u}_0\|_{\dot H^{1/2}} \le \varepsilon\nu$
Pillar III (BKM)	One-directional	$\int_0^T \|\boldsymbol{\omega}\|_{L^\infty}\,dt < \infty \implies T^* > T$
Clay Statement (B)	Open	Large-data global regularity

References

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[3] J. T. Beale, T. Kato, and A. Majda, Remarks on the breakdown of smooth solutions for the 3-D Euler equations, Comm. Math. Phys. **94** (1984), 61–66.

[4] H. Kozono and Y. Taniuchi, Bilinear estimates in BMO and the Navier–Stokes equations, Math. Z. **235** (2000), 173–194.